

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Summer Examination – 2025**

Course: B. Tech Branch: Mechanical Engineering / Mechanical Engineering (Sandwich) Semester: VIII

Subject Code & Name: Fundamentals of Automotive Systems BTMEC802A**Max Marks: 60****Date: 03/07/2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
	1. The major loss in a S.I. engine is due to _____ a) variation in specific heat and chemical equilibrium b) pumping c) exhaust blow down d) incomplete combustion 2. Which of the following is the most preferred S.I. engines fuel? a) Napthenes b) olefins c) aromatics d) paraffins 3. Which of the following statement is not correct with respect to alcohols as alternate fuel in IC engines? a) alcohol contains about half the heat energy of gasoline/litre b) alcohol does not vaporize as easily as gasoline c) alcohols are corrosive in nature d) anti-knock characteristics of alcohol is poor 4. Which of the following does not relate to spark ignition engine? a) Spark plug b) Carburetor c) Fuel injector d) Ignition coil 5. If there are 7 clutch plates in a multi-plate clutch, what is the number of pair of contact surfaces? a) 5 b) 4 c) 6 d) 8		1 mark each

	6. Which of the following contains no linkage between the clutch and the pedal? a) Clutch – by – wire b) Wet clutch c) Hydraulic single plate clutch d) Hydraulic multi-plate clutch 7. In which type of manual transmission the double-declutching is used? a) Constant-mesh gearbox b) Sliding mesh gearbox c) Synchromesh gearbox d) Epicyclical gearbox 8. Where is the overdrive located? a) Between transmission and engine b) Between transmission and rear axle c) Between transmission and propeller shaft d) Between transmission and differential 9. What is the angle between the steering axis and the vertical in the plane of the wheel? a) Castor b) Camber c) Steering axis inclination d) Kingpin inclination 10. What is a condition called when the vehicle will try to move away from its normal direction and to keep it on the right path there is need to steer a little? a) Understeer b) Oversteer c) Reversibility d) Irreversibility 11. Generally which brakes are on the front wheels? a) Drum brake b) Disk brake c) Shoe brake d) Double shoe brake 12. Which type of wheels is preferred in sports cars? a) Disc wheel b) Wire wheel c) Magnesium alloy wheel d) Aluminium alloy wheel		
Q. 2	Solve the following.		
A)	Explain the components of SI Engine with suitable schematic diagram.	CO1	6
B)	What is supercharging? How it is done explain with necessary sketch.	CO1	6
Q.3	Solve the following.		12
A)	With neat sketches explain different phases of Combustion in CI Engine.	CO1	6
B)	What is fuel injection systems? Enlist types of injection systems and explain.	CO2	6

Q. 4	Solve Any Two of the following.		12
A)	What is Double declutching? How this problem is solved in Synchromesh gear box design.	CO2, CO4	6
B)	What is a Clutch? Explain Construction and working of Single Plate and Multi plate clutch with sketches.	CO1	6
C)	Derive the expression Tractive force available at tyre road interface in the powertrain analysis.	CO2	6
Q.5	Solve Any Two of the following.		12
A)	With neat sketch explain working of internal expanding brake and enlist its advantages.	CO1	6
B)	What are the emission control systems used to control pollution in automobiles? Explain.	CO5	6
C)	What is ABS? Discuss with a neat sketch the operation Methodology of ABS.	CO5	6
Q. 6	Solve Any Two of the following.		12
A)	Explain with neat sketch working of hydraulic power steering system.	CO2	6
B)	With neat sketch explain working of twin tube shock absorber.	CO2, CO3	6
C)	Differentiate between Tyres with tube and Tubeless tyres.	CO2, CO3	6
	*** End ***		